HYQAD: Hybrid AI-Quantum Anomaly Detection for QKD Systems

Cybersecurity Research Internship — Progress Report

Arnav Kapoor

July 2026

CCRI Lab

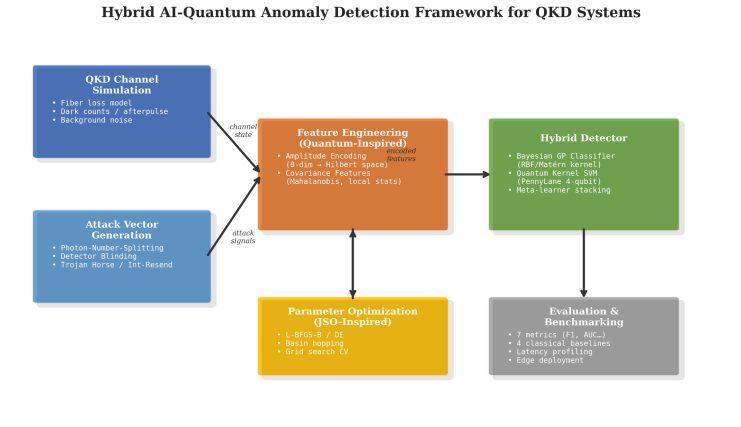
Project Overview & Objectives

HYQAD: Hybrid AI-Quantum anomaly detection framework for Quantum Key Distribution (QKD) systems

- Detect physical-layer attacks (PNS, Detector Blinding, Trojan Horse, Intercept-Resend)
- Bayesian GP classifier providing **calibrated uncertainty** estimates
- 3-qubit quantum kernel SVM + quantum-inspired feature engineering (amplitude encoding, MCD covariance)
- Sub-10ms edge-deployable inference

Completed: Framework implementation ($\sim$2000 loc), 5 research articles, 30 unit tests, full benchmark suite

Architecture



Benchmark Performance (5000 samples, 28-dim features)

Metric	RF	Hybrid	OC-SVM	IF	LOF
Acc.	**0.995**	0.940	0.894	0.748	0.755
F1	**0.992**	0.889	0.810	0.269	0.302
AUC	**1.000**	1.000	0.888	0.999	0.905
MCC	**0.987**	0.839	0.746	0.413	0.437
Spec.	0.996	**0.999**	0.958	0.594	0.628

- RF dominates point estimates; HYQAD uniquely provides **calibrated uncertainty**
- RF near-perfection reflects **synthetic data difficulty** – real-world validation needed

Ablation Study & Key Findings

Ablation (mean $\pm$ std, 3 seeds):

Metric	RF	GP-only	QK-only	Hybrid
Acc.	**1.000**	$0.900_{\pm 0.042}$	$0.700_{\pm 0.000}$	$0.782_{\pm 0.077}$
F1	**1.000**	$0.795_{\pm 0.098}$	$0.000_{\pm 0.000}$	$0.384_{\pm 0.347}$

- GP drives accuracy; quantum kernel (25 training samples) does not yet contribute positively
- $\mathcal{O}(n^2)$ circuit cost limits QK training size – hardware acceleration needed

Uncertainty calibration: Rejection-based deferral ($u > 0.4$) raises accuracy to 0.982 (86.3% coverage)

Latency Profile (Intel i7, CPU)

Component	Latency (ms)	vs. RF
RF (200 trees)	0.9	1.0×
Bayesian GP	5.8	6.4×
Quantum kernel SVM (offline)	50.0	55.6×
Full pipeline (Bayesian)	**8.5**	**9.4**×
Bayesian linear est.	**0.4**	**0.4**×

- Bayesian linear estimator: 0.4ms – suitable for real-time edge deployment
- Quantum kernel is the bottleneck; approximations (random Fourier features) planned

Challenges & Planned Work

Challenges:

- Quantum kernel latency (50ms/sample on simulator) – exploring Nyström/random Fourier features
- Synthetic data may be too easy (RF AUC = 1.000) – experimental validation on physical testbeds needed
- Attack parameters are modeling assumptions – need calibration from real QKD hardware data

Planned milestones:

Week 3	Stress-test high-noise regimes, extend attack library
Week 4	FPGA feasibility study, experimental validation, paper draft
Month 2	Adaptive GP thresholding, FPGA implementation, real hardware benchmarks

Thank You

Questions?

HYQAD: Hybrid AI-Quantum Anomaly Detection for QKD Systems